NHS data analysis report

Introduction:

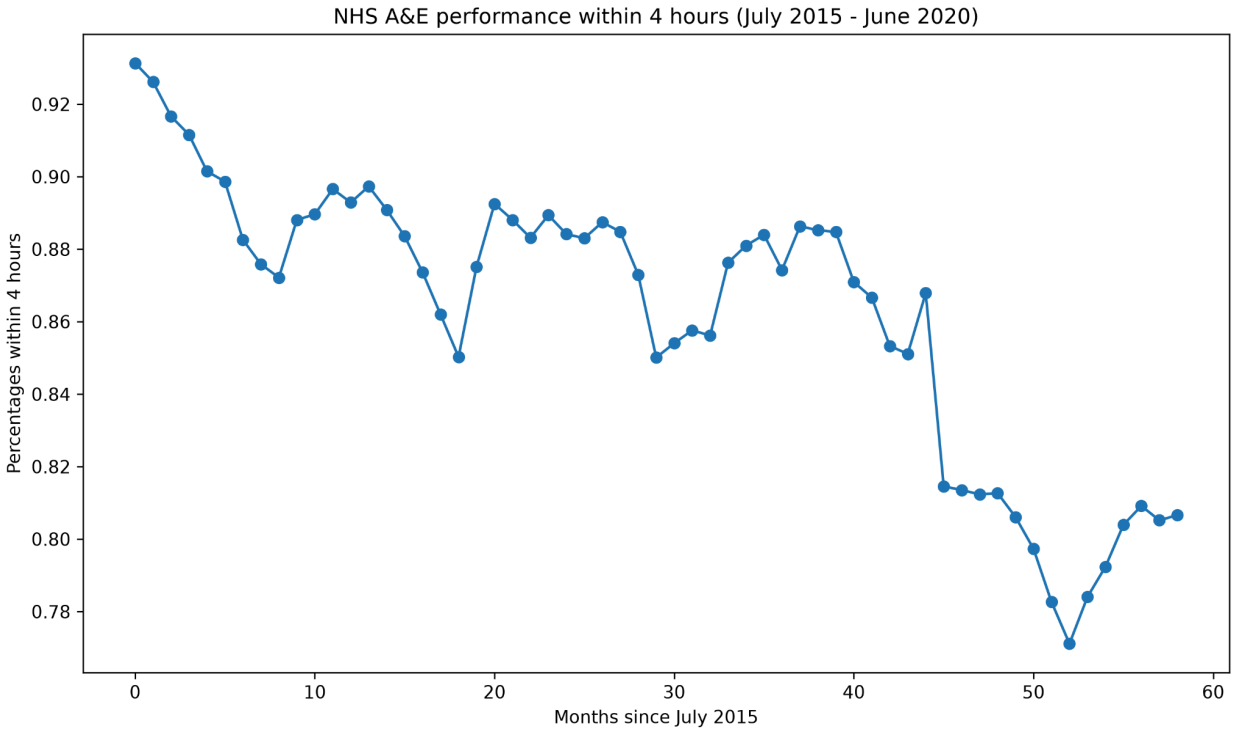
This report aims to analyse whether the NHS and A&E performance against the 4 hour target has seen a decline across trusts between July 2015 to June 2020 using A&E attendances England data set by Treich (Kaggle). As stated by Jefferies et al. (2025) the NHS has seen reduced public satisfaction, with 73% of respondents being dissatisfied with NHS waiting times. Therefore, analysing wait time across the UK may help to illustrate where the wait times are at their worst.

Methodology:

- Used Percentage in 4 hours or less (all), averaged across all NHS trusts to one national figure per month.
- Dropped all months pre-July 2015 as earlier values in this column had inconsistent scaling.
- Excluded April 2019, as every NHS trust's values were invalid (mean = 1.72 for a 0 -1 fraction).
- Applied 0 - 1 validity filter as a safety net, removed all the duplicates and nulls.
- Ran a linear line regression of the monthly percentage against time (since July 2015) using scippy.stats.lineregress.

Findings:

3a: NHS A&E Performance within 4 hours (July 2015 to June 2020)



- As shown by 3a, the NHS saw a performance drop from around 93 percent to around 80 percent during this span of time. The slope = -0.00191 (about 0.19 percentage loss per month, or around 2.3 per year). This suggests that there is a consistent decline in performance.
- The strength: r = -0.845, so decline is steady.
- The regression gave a significance value of $p < 0.001$.
- All the findings suggest that the NHS performance decline is real and has been consistent over the period of time. Which coincides with the report from Jefferies et al. (2025) referring to NHS waiting time dissatisfaction. However, this analysis is only descriptive: it establishes whether and how strongly performance has declined, but the dataset corinthians no real information on potential reasons for the causes of the decline. Therefore, explaining causes would need further research.

Limitations:

- The national figure is an unweighted average between all trusts, so it does not account for the size difference between trusts. A small one is treated the same as a large one.
- The data included specialist trusts (eg, cancer and maternity) with a placeholder 0, which pulls the level down. The trend is not finding the exact level.
- Some of the data overlaps with Covid-19, which can explain the changes in the pattern.

- Two data- quality exclusions were needed (pre-July 2015, April 2019)

Conclusion:

In conclusion, the NHS and A&E do seem to have a statistically significant and steady decline of the performance against the 4 hour target. However, with the lack of deeper level research into the causes, a reason for the decline cannot be derived.